

RESEARCH

Open Access



Deep learning-driven automatic music score recognition and digital transcription algorithm

Huina Li^{1*} and Xinyan Song²

*Correspondence:

Huina Li

huina_nana@outlook.com

¹School of Art, Jiaozuo University,
Jiaozuo 454000, Henan, China

²School of International,
Zhengzhou University,
Zhengzhou 450001, Henan, China

Abstract

Automatic music score recognition, also known as Optical Music Recognition (OMR), is required for transforming printed or handwritten sheet music to a digital, editable version. Traditional rule-based OMR systems often fail under complex staff layouts, noise, and stylistic variations, resulting in reduced transcription reliability. This research proposes a deep learning framework named Transformer-based Convo Memory Network optimized with Locust Swarm Algorithm (TCMN-LSA) to enhance recognition accuracy, convergence stability, and robustness across diverse music manuscripts. The model is trained and evaluated using the Sheet Music Transformer dataset, which includes the GrandStaff and Camera-GrandStaff subsets containing real-world distortions. Preprocessing improves image consistency using Z-score normalization and Wiener filtering. The TCMN-LSA architecture integrates a CNN feature extractor for spatial symbol detection, a Transformer encoder to model contextual dependencies, and an LSTM module for temporal sequence learning. A Transformer-based decoder produces structured symbolic notation. The Locust Swarm Algorithm (LSA) performs adaptive optimization, improving convergence efficiency and reducing parameter instability. Connectionist Temporal Classification (CTC) enables alignment-free transcription for variable-length polyphonic inputs. Experiments are performed using Python, TensorFlow, and PyTorch, ensuring high-performance training and large-scale batch processing. Evaluation metrics include precision, recall, accuracy, CER, SER, and LER. The proposed model achieves 95.84% of precision, 82.67% of recall, and 88.41% of accuracy, significantly outperforming classical approaches. On the GrandStaff dataset, TCMN-LSA attains 1.62% CER, 2.10% SER, and 6.45% LER, and 1.21% CER_{bug}, demonstrating improved symbol recognition and transcription stability. These findings confirm TCMN-LSA as a scalable and reliable solution for music archiving, score editing, and educational applications.

Article Highlights

- Proposes TCMN-LSA for OMR, fusing CNN/Transformer/LSTM with LSA optimization to enhance recognition accuracy and distortion robustness.
- Integrates preprocessing and CTC loss, enabling alignment-free polyphonic transcription with stable convergence for OMR.
- Achieves 95.84% precision, 88.41% accuracy and low error rates, outperforming classical OMR methods significantly.



Keywords Music education, Optical music recognition (OMR), Music score, Digital, Deep learning (DL)

1 Introduction

A multi-resolution prediction framework based on interpretation of musical structures based on sheet images and real-time score image tracking is advanced [1], which aims to reduce latency, increase alignment accuracy, and deal with the challenges in dynamic musical passages, noisy symbols, and complex layouts in notation that occur during ongoing music-following tasks. Deep Neural Networks (DNNs) are used to read the organ tablature notation automatically based on the patterns of symbols and structural relationships [2], which overcomes the limitations of the rule-based principles of decoding, enables dense polyphonic textures, and improves the recognition rate of historical tablatures with extremely variable glyph shapes and non-standard layouts. Automatic translation procedures are proposed to redefine OMR encoding, which allows more adaptable encoding of visual notation to machine-readable sequences [3], and extends the generalization behaviours across the different music forms, and enhances the ability of end-to-end encoding to handle noisy, handwritten, and complex document sensitivities.

The CTC-based learning is applied to learn chroma features for the accurate score-audio retrieval without explicit alignment [4]. The model develops the temporal-harmonic patterns directly based on the data, overcoming variability in tempo, timbre, and performance style, to provide reliable matching of symbolic notation and the related audio recording. The multimodal scheme of visual sheet images and audio cues enhances the accuracy of the music transcription process [5]. By integrating cross-modal representations that increase the strength in the recognition of symbols, clear up ambiguities in the complex notation, and reduce errors that normally occur when cues are used solely to identify symbols. An algorithmic digital signal processing pipeline is built to achieve music transcription on machine learning models to identify pitch, onset, and structure [6], which is more responsive and provides more accurate features based on noisy audio, and could deal with limitations of the traditional frequency-based methods in fast or polyphonic musical content.

Artificial intelligence (AI) techniques are used to transcribe Tatar folk songs, adhering to the pitch contours, melody structure, and ornamentation of the style of folk songs [7]. It was intended to preserve cultural heritage by reconciling inconsistencies during manual transcription and to resolve problems of microtonality and non-Western melodic phrasing. The multimodal piano music identification method applies marketplace fingerprinting to correlate performances with large-scale databases [8] by matching symbolic structures, audio features, and metadata that solve variability in playing style and recording quality, and increases retrieval performance in large music collections.

The object detection models are adapted to identify white mensural notation and translate it into modern notation [9]. It was effective in repairing unusual symbols used during history, odd spacing, or damaged prints, which are higher than rule-based methods in restoring rhythmic values and structural validity in the early-music documents. AI-based processing assists the recognition of Hungarian folk-song transcriptions written by hand, learning style pen pattern, and notational variability [10], enhancing

consistency of symbols, decreasing the amount of work on manual correcting, and adapting to the characteristics of handwritten documents, including variable spacing and unclear stroke formations. A decoupled OMR system decouples notation modeling and visual recognition, allowing flexible end-to-end learning [11]. The implementation addresses the error propagation issue, increases the scalability of the system to notation types, and the sequence reconstruction of symbols, particularly with dense and polyphonic or visually rich score representations. The end-to-end neural OMR is introduced with monophonic scores instead of using handcrafted pipelines, with joint learning of symbol shapes and temporal order [12], yielding a stronger recognition of A4 non-sniffed sheet images, requiring less dependence on preprocessing, and removing the limitation of segmentation-based methods to distorted or noisy sheet images.

The TCMN-LSA model proposed is effective in identifying piano music scores as it combines space-temporal deep learning with the adaptive optimization of a swarm. The major research key contributions are as follows:

- Transformer-based Convolutional Memory Network (TCMN): CNN extracts rich spatial patterns, Transformer encoders capture long-range dependencies, Long Short-term Memory (LSTM) preserves temporal continuity, and the Transformer decoder converts structured musical notation.
- Locust Swarm Algorithm (LSA): Adaptively optimizes key parameters, stabilizes convergence, feature-selection efficiency, and robustness over a range of musical notations, with a CTC mechanism to maintain alignment-free transcription.
- The suggested method proves to be significantly better than the current practices, providing better and more credible results without having to cite any performance measurements.

2 Related works

The recent trends in Optical Music Recognition (OMR) and music transcription show that a strong reliance on convolutional neural networks (CNNs), transformer-based, and hybrid deep-learning models primarily contributes to the improved performance of the systems in terms of symbol recognition, pitch estimation, and the structural understanding of musical scores.

An OMR system based on the multi-kernel CNNs and synthetic music augmentation was designed to handle homophonic scores. Homophonic experimentation found a symbol-level accuracy of between 93% and 95%, with better note duration and rest detection. The model was a step further in defeating the baseline CNNs by 2 to 4%. However, the performance dropped below 80% in the case of polyphonic inputs, and the system showed challenging behavior in using handwritten pages or noisy pages [13].

Another note-level singing melody transcription system based on transformers used self-attention models to capture time. On singing datasets, the accuracy of the evaluations was reported at 89% and 92%, respectively, in pitch and voicing accuracy, and thus outperformed LSTM-based baselines by 6% and 8%, respectively. The procedure, however, required a large amount of labeled datasets, and accuracy dropped by 10 to 15% when using noisy or vibrato-heavy singing conditions [14].

An OMR monophonic camera-based score that used a dual-branch network topology to recognise staves and symbols was a staff-aware OMR. These findings showed that the detection rate of staves was 97% with a symbol recall of 90%, and the number of false

positives was lower than with the conventional CNN models. However, the performance was significantly reduced (to $\sim 75\%$) with rotated or perspective-distorted images, and so robustness was restricted to variability of mobile capture [15].

The Multilevel Cascade Residual Recurrent (MCRR) model is a model that fused CNN and former recurrent residual blocks to decode musical symbols. The numeric results obtained gave a symbol classification system of 96% and a sequence error cut of 12% compared to non-cascade models. The ablation experiments proved that consistent gains were made at all levels; however, the model was computationally demanding, and the inference time was nearly twice as much as the baseline CRNNs [16].

The transcription model was a multimodal model, which combined audio-lyric alignment by cross-modal attention. It obtained an alignment precision of more than 94%, and compared with audio-only baselines, the word error rate (WER) was decreased to 11%. However, there was a reduction of close to 20% in performance on non-studio records, and the procedure relied greatly on paired audio text records [17].

The medieval staves, neumes, and textual elements were identified in a complete pipeline of medieval chant OMR. On historical manuscripts, the model had a neume recognition error of about 90% and a staff detection error of more than 95%. The major weakness was that it could not generalize to modern notation (below 70% accuracy) and used manually annotated medieval datasets [18].

Table 1 shows the overview of the aims, methods, results, and limitations of recent OMR approaches.

A two-dimensional agnostic model of notation used a CNN-based spatial analysis to identify musical symbols without considering staff information. It was reported that the numerical evaluations had a precision of 93% and a recall of 91%, which was a 5 to 7% improvement compared to the traditional rule-based systems [25]. Nevertheless, the model was poor in handwritten or multi-voice scores, with an accuracy of almost 75%.

A sophisticated OMR system was built in with pre-processing, segmentation, and deep classification functions of the contemporary printed scores. According to experiments, the accuracy of segmentation was around 95%, and the accuracy of the symbol classification was about 92%, and there was consistent performance when the imaging was

Table 1 Comparison of recent optical music recognition methods

Ref.	Aim	Methods	Results	Limitations
[19]	Standardized OMR evaluation	Unified scoring framework	Improved cross-system comparability; variance in metric interpretation reduced by $\sim 30\%$	Does not enhance recognition accuracy; only supports evaluation
[20]	Tonal representation from scores	Deep pitch-class embeddings	Tonal-correlation score ≈ 0.92 , outperforming classical pitch histograms	Limited to tonal structure; cannot recognize symbols
[21]	End-to-end full-page OMR	Layout-aware transformer	Page-level detection accuracy 94%, symbol accuracy 88%	Drops to $\sim 70\%$ on handwritten pages
[22]	Piano audio transcription	CNN spectral model	Note-onset F1-score 87%, transcription accuracy $\sim 82\%$	Weak on fast passages; harmonic overlap reduces accuracy by 10–12%
[23]	Staff-region detection with uncertainty	Monte Carlo dropout CNN	Detection accuracy 96%, uncertainty reduction $\approx 15\%$	Expensive sampling increases computation 5–10x
[24]	Handwritten symbol recognition	Ensemble of transfer-learning models	Classification accuracy 94–96% across datasets	Inference time is nearly 3x slower due to the model ensemble

controlled. However, the accuracy of recognition became 12 to 18% lower on curved, shadowed, or low-resolution score pages, which limited its application to the real world [26].

An instrument recognition model based on a CNN model extracted spectral features that were used to identify a timbre belonging to an instrument family. The system has a total accuracy of 93% with a precision and recall of over 90% on most classes. Even though it is not purely OMR-specific, its acoustic feature knowledge can be used to supplement multimodal Automatic Music Transcription (AMT) pipelines. Its weakness was in its poor results (around 78%) in polyphonic mixes and blended timbers, limiting its use to an ensemble setting [27].

The auditory filter-bank-based method was used for the transcription of polyphonic music through the modelling of human cochlear frequency filtering. The experiments showed that multi-pitch detection was improved by 6 to 8% and that there was better distinction between harmonics that overlapped with each other. Nonetheless, thick chords with powerful spectral masking impaired accuracy and thus, performance in highly polyphonic passages was weakened [28].

A true end-to-end OMR system of piano-like scores using CNN-Transformer (CNNT) architecture was able to recognize symbols and structural layout together. The model had a symbol accuracy of about 92% and a better staff-line handling [29], but abnormal spacing and poor quality scans led to a fall in the accuracy of about 12%.

An effective pipeline of feature-extraction and sequence-decoding was applied to practical piano-form OMR, and optimized pre-processing as well as structured decoding was used. The technique acquired a 94% page-level accuracy at lower inference time. However, with handwritten or distorted score pages, the performance decline became observable, hence restricting the performance strength in the various types of notations [30].

2.1 Research gap

Despite the recent developments of Optical Music Recognition studies that have shown significant advancements with CNNs, transformers, and hybrid deep-learning models, a number of limitations still exist. CNN-based approaches provide good performance at the symbol-level recognition but are unable to handle long-range dependencies in the context and deep polyphonic features. Transformer methods further improve global context modeling, yet often have poor performance with distorted or handwritten scores and have very high needs for annotated data. Multimodal and end-to-end models enhance the quality of transcription, but create high computational complexity and sensitivity to alignment. In addition, the majority of the studies found in the literature are based on fixed optimisation strategies, hence limiting the stability of convergence in different notation styles. These constraints explain the need to have a robust architecture that includes a combination of spatial, temporal, and contextual information, and still has the ability to optimise the structure. The suggested TCMN-LSA framework tackles these issues and combines CNN-based spatial modelling with Transformer-LSTM temporal learning and adaptive swarm-based optimisation to allow accurate and alignment-free transcription on clean and real-world datasets.

3 Methodology

The approach uses a pipeline structure that is designed to transcribe music notation of piano-like music accurately with the help of the Sheet Music Transformer Dataset. The pre-processing includes Z-score normalisation to have a uniform input scale and a Wiener filter to reduce noise and improve the clarity of symbols. The TCMN-LSA model uses CNNs to extract local features, a Transformer encoder to extract global notation context, and an LSTM to extract the sequential flow of symbols. The decoder produces output final symbols, and LSA is used to adjust the parameters to optimise the recognition performance, and CTC allows transcription of continuous music sequences without alignment. The entire methodological process of the proposed TCMN-LSA model is illustrated in Fig. 1.

3.1 Data collection

The Sheet Music Transformer Datasets (PRAIG/GrandStaff and PRAIG/Camera-GrandStaff - <https://huggingface.co/collections/antoniorv6/sheet-music-transformer-datasets>) support musical-symbol recognition and piano-form transcription. The GrandStaff test data consists of 53,882 digitized staff images having correct symbolic labels, thus supplying clean and manipulated inputs to design effective OMR models. The Camera-GrandStaff dataset consists of 53,882 camera-captured images that exhibit real-world distortions like blur, light variation, warping, and perspective noise, and thus, represent real-world conditions of mobile capture. These datasets provide balanced, diverse, and complementary samples of sheet music and thus guarantee an overall assessment of the model operation in both ideal and real-environment conditions without distorting the initial content and meaning. Sample piano sheet music is given in Fig. 2.

3.2 Data pre-processing for using clean and consistent

Before recognition, standardisation of the intensity levels, through Z-score normalization, is applied, followed by a Wiener filter to remove noise and blur.

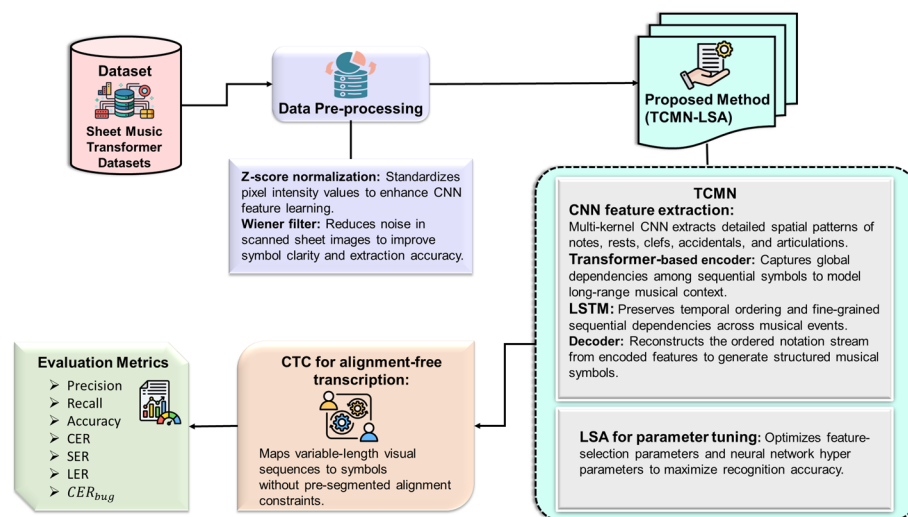


Fig. 1 Workflow of the TCMN-LSA model

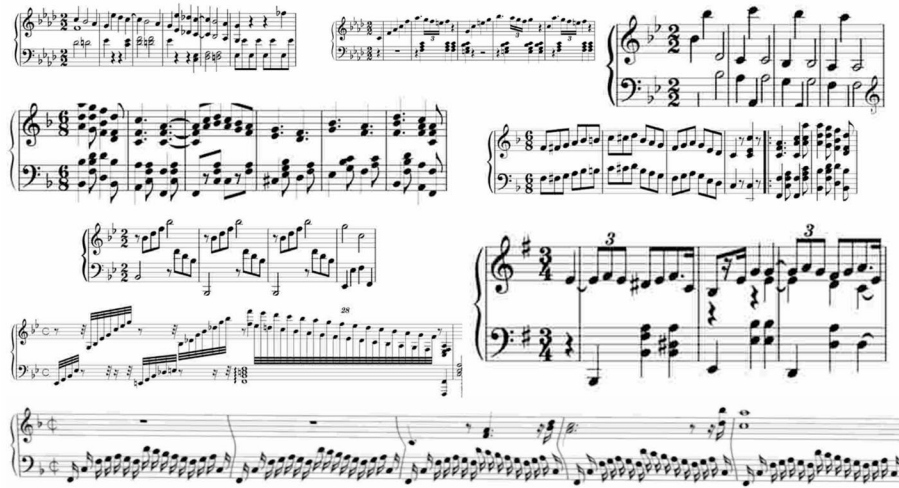


Fig. 2 Sample piano music sheets music for both Grandstaff and Camera Grandstaff

3.2.1 Z-score normalization for consistency and readability

The main preprocessing step involves Z-score intensity normalisation to improve visual consistency in piano-sheet-music images, followed by deep -learning feature extraction and symbol recognition. Piano scores have a high frequency of non-uniform illumination, scanning artefact, blurring staff lines, and uneven contrast between notation symbols and background. To standardize these variations, each pixel intensity Z_t is normalized and used in Eq. (1)

$$Z_t = \frac{Z_t - \mu}{\sigma} \quad (1)$$

Where Z_t represents the pixel intensity at location t , μ is the mean intensity of the entire image, and σ is the corresponding standard deviation. This standardisation enhances minor musical elements, including note heads, lines of bars, beams, and areas of staff, by reducing page illumination bias. With the use of a single and statistically based normalisation step, consistency is achieved among different sheets of piano music, leading to better consistency in the following segmentation and symbol-recognition modules. The sophisticated preprocessing is thereby helpful in the extraction of features that are more reliable and enhances the overall performance of music-image analysis.

3.2.2 Wiener filter using clarity of the input sheet music

The reliability of deep learning-based Optical Music Recognition heavily depends on the clarity of the input sheet-music images. It applies a Wiener-filter-based noise reduction method during preprocessing. Sheet-music scans often contain speckle noise, ink bleed, uneven illumination, and background texture variations that distort staff lines and musical symbols. The noisy pixel intensity $w(m)$ value at location m in the piano sheet-music input image is modelled in Eq. (2)

$$w(m) = t(m) + g(m) \quad (2)$$

Where, $t(m)$ represents the clean music-score image, and $g(m)$ denotes additive noise, m is the pixel location index. The Wiener-filtered output $\hat{t}(m)$ is computed through optimal linear filtering used in Eq. (3)

$$\hat{t}(m) = \sum_{l=0}^{M-1} i(l) w(m-l) \quad (3)$$

Where $i(l)$ denotes the Wiener filter coefficient applied at position for noise suppression, and M is the filter size, l represents the filter position within the convolution window, and $w(m-l)$ is the neighboring noisy pixel intensity. Noise reduction carried out in the frequency domain is expressed in Eq. (4)

$$\hat{T}(\omega) = I(\omega) W(\omega) \quad (4)$$

Where $\hat{T}(\omega)$ is the filtered image in the frequency domain, $W(\omega)$ is the noisy image spectrum, and $I(\omega)$ is the Wiener filter function, ω is the angular frequency variable. The Wiener-Hopf formulation defines the optimal filter as Eq. (5)

$$I(\omega) = \frac{P_{TT}(\omega)}{P_{TT}(\omega) + P_{WW}(\omega)} \quad (5)$$

Where, P_{TT} and P_{WW} represent the power spectral densities of the clean image and noise. Moreover, the preprocessing phase improves staff-line sharpness and the visibility of the symbols, thus providing accurate feature extraction and leading to strong and high-precision automatic music-score recognition.

3.2.3 Transformer-based convo memory network with locust swarm algorithm (TCMN-LSA) for music score recognition and digital transcription

The TCMN-LSA architecture has been chosen to deal with the intrinsic spatial, contextual, and temporal complexity of piano-sheet-music-recognition. CNNs are useful in capturing local visual patterns such as note heads and staff lines, and the Transformer model captures long-range contextual relationships between measures. Transformer-only models, however, can fail to capture fine-grained temporal continuity; an LSTM is therefore added with the express purpose of preserving rhythmic order and sequence of symbols. The hybrid architecture can have a complex non-convex optimisation landscape, where traditional optimisers can find a sub-optimal solution. The LSA is hence used to improve the stability of convergence and parameter exploration, which generated better transcription precision and strength in alignment-free music-score recognition.

3.2.4 Transformer-based convo memory network (TCMN)

A single TCMN-based recognition system is used to transform the image of piano sheet music into structured symbolic notation. This starts with a CNN feature extractor, which marks musical primitives- notes, rests, clefs, and accidentals, as well as articulations. These visual elements are passed on to a Transformer encoder, which builds global dependencies, and an LSTM module that builds sequential and contextual transfers among the identified symbols. The transformer decoder recreates the ordered representation stream, and a CTC layer provides alignment-free transcription of the visual sequences through the transformation of variable-length visual sequences to outputs of symbols.

3.2.5 Convolutional neural network (CNN)-based spatial feature extraction module

A multi-kernel CNN module is used to extract rich spatial features of a piano-sheet-music image; symbols used in piano-sheet-music are not uniform and differ in shape,

size, and thickness of the stroke. Parallel convolutional kernels can capture the staff-line geometry, note-head contours, and fine-grained notation features, thus giving a detailed representation. The input image sequence is represented as Eq. (6)

$$\tilde{Y} = \left(\tilde{Y}_1, \tilde{Y}_2, \dots, \tilde{Y}_{u_x} \right)^U \quad (6)$$

Here, $\tilde{Y}$ is input image sequences for piano sheet-music, representing all time slices, $\tilde{Y}_1, \tilde{Y}_2, \dots, \tilde{Y}_{u_x}$ represents individual time-step images or slices extracted from the input sequence, each slice contains staff lines, notes, and other musical symbols, U is transpose operator, indicating that the sequence is arranged along the temporal dimension for preprocessing, with dimensionality (u_x, I, X) , where u_x is the temporal slice count, I is image height, and X is image width. After multi-kernel convolution, the output becomes $(u_x, d, I - 1, 1)$, where d denotes the number of feature channels. After removing redundant dimensions, the final CNN feature representation is expressed as $(u_x, I - 1)$. This structure ensures that each time slice is processed independently, preserving symbol-level ordering. The extracted spatial patterns, notes, rests, clefs, accidentals, and articulations form the foundational input for the encoder, LSTM sequence model, and Transformer-based notation decoder.

3.2.6 Transformer encoder-based temporal feature extraction module

In the piano sheet-music recognition pipeline, the Transformer encoder is employed to extract temporal and contextual features from sequential CNN-extracted spatial representations. After CNN processing, the feature map of dimension $(u_x, I - 1)$ is positionally encoded to retain the ordering information of symbols along the temporal axis. A dimension transformation layer linearly maps these low-dimensional features into a higher-dimensional space of size (u_x) to match the Transformer's internal representation. The combined positional and transformed vectors, $\hat{Y} \in \mathbb{R}^{u_x \times e_{model}}$, serve as the input for the multi-head attention layer. The multi-head attention mechanism independently projects $\hat{Y}$ into queries, keys, and values for each head, as represented in Eq. (7)

$$R_i = \hat{Y}^{u_x \times e_{model}} X_i^R, L_i = \hat{Y}^{u_x \times e_{model}} X_i^L, W_i = \hat{Y}^{u_x \times e_{model}} X_i^W, i = 1, 2, \dots, I, \quad (7)$$

Where, $\hat{Y}$ is input feature sequence to the Transformer encoder, output from CNN, dimension (u_x, e_{model}) , u_x is the number of temporal slices or time steps in the CNN feature sequence, e_{model} is the dimension of encoded feature vector in the transformer, R_i, L_i, W_i denotes the Query, key, and value matrices for the i^{th} attention head, representing linear projections of $\hat{Y}$, X_i^R, X_i^L, X_i^W are the trainable linear projection matrices for queries, keys, and values of the i^{th} head, I is number of attention heads in the multi-head attention layer. The attention output for each head is computed using scaled dot-product attention, as expressed in Eq. (8)

$$context_i = \text{Attention}(R_i, L_i, W_i) = \text{softmax} \left(\frac{R_i L_i^U}{\sqrt{e_l}} \right) W_i \quad (8)$$

Here, $context_i$ is the output of scaled dot-product attention for the i^{th} head, softmax is normalization function, L_i^U key matrix multiplied by query vector transpose, $\sqrt{e_l}$ is a scaling factor, e_l is the dimension of each attention head, calculated as $e_l = e_{model}/I$,

All head outputs are concatenated and linearly mapped to produce the final attention vector, which is defined as Eq. (9)

$$Y_{atten} = \text{concat}(context_1, \dots, context_I) X_P \quad (9)$$

Here, Y_{atten} denotes concatenated multi-head attention output, dimension (u_x, e_{model}) , $context_1, \dots, context_I$ is the output of scaled dot-product attention for the I^{th} head, X_P is a trainable linear mapping matrix for combining multi-head attention outputs. Residual connections and normalization are applied, followed by a feed-forward layer with ReLU activation, expressed in Eqs. (10 & 11)

$$ff = an(fc_2(ReLU(fc_1(an(Y_{atten})))))) \quad (10)$$

$$ReLU(y) = \max(0, y) \quad (11)$$

Where, ff is the output of the feed-forward layer after $ReLU(y)$ and normalization, $an(.)$ denotes the Add & Norm operation (residual connection followed by layer normalization), and fc_1, fc_2 are fully connected layers in the feed-forward network. The output of the feed-forward layer (opt) passes through an LSTM and an Add & Norm layer to produce the Encoder output in Eq. (12)

$$opt = an(LSTM(ff)) \quad (12)$$

The final encoder output after $LSTM$ and an additional layer, representing a temporally and contextually enriched feature sequence for the decoder. Technically, the Transformer encoder in this research captures long-range dependencies between musical symbols, enabling recognition of patterns across the piano score. The integration with LSTM preserves sequential correlations, ensuring that symbol context, such as simultaneous notes, articulations, and rhythmic structures, is accurately encoded for the subsequent Transformer decoder and CTC-based transcription.

3.2.7 LSTM is used to capture temporal dependencies and sequential patterns in musical symbol sequences

The output of the Transformer encoder is further processed by an LSTM layer to enhance the extraction of temporal dependencies that are essential in musical sequences. While the Transformer captures global contextual relationships through self-attention and positional encoding, musical structures such as rhythmic continuity, note duration, and melodic progression are also considered.

Although Transformer encoders capture long-range contextual relationships via self-attention, they do not explicitly model sequential state continuity. In music-score recognition, precise temporal ordering and rhythmic consistency are essential, especially for polyphonic sequences. The LSTM layer complements the Transformer by explicitly modeling temporal transitions through gated memory, preserving ordering-sensitive dependencies and stabilizing symbol progression. This hybrid design reduces sequence-level errors, as evidenced by lower SER and LER values, justifying the inclusion of LSTM after the Transformer.

The LSTM consists of three gates: the forget gate ft_g , input gate it_g , and output gate ot_g , which control how temporal information flows through the network, as shown in Eqs. (13–15).

$$ft_g = \sigma (M_{ft} \cdot [p_{g-1}, z_g] + d_{ft}) \quad (13)$$

$$it_g = \sigma (M_{it} \cdot [p_{g-1}, z_g] + d_{it}) \quad (14)$$

$$ot_g = \sigma (M_{ot} \cdot [p_{g-1}, z_g] + d_{ot}) \quad (15)$$

Where, σ is a sigmoid activation function, M_{ft} , M_{it} , M_{ot} are trainable weight matrices, p_{g-1} is the hidden state (external output) from the previous time step, carries temporal context, z_g is the input vector at the current time step (CNN+Transformer encoder features for the current note slice), d_{ft}, d_{it}, d_{ot} is trainable bias vectors. The candidate state ($\tilde{dt}_g$), memory cell (dt_g), and output state (p_g) are computed in Eqs. (16–18)

$$\tilde{dt}_g = \tanh (M_{dt} \cdot [p_{g-1}, z_g] + d_{dt}) \quad (16)$$

$$dt_g = ft_g \cdot dt_{g-1} + it_g \cdot \tilde{dt}_g \quad (17)$$

$$p_g = ot_g \cdot \tanh (dt_g) \quad (18)$$

Here, $\tanh$ is the hyperbolic tangent activation function, M_{dt} , d_{dt} are the weight matrix and bias vector of the cell state, and dt_{g-1} is the cell state from the previous time step. The activation functions used within the LSTM gates remain: Sigmoid function ($\sigma(z) = \frac{1}{1+e^{-z}}$), and hyperbolic tangent function ($\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$), where $\sigma(z)$ is the sigmoid function controlling LSTM gate activations for note-sequence memory, e^z , $-e^z$ is exponential terms used in σ and $\tanh$ functions to regulate temporal scaling. In music-score recognition, these equations allow the LSTM to retain essential musical patterns (e.g., recurring motifs), forget irrelevant transitions, and output temporally consistent representations. This strengthens temporal modelling, improving pitch-sequence decoding, rhythm alignment, and recognition accuracy for complex musical passages.

3.2.8 Transformer decoder-based symbol sequence reconstruction module

In the piano sheet-music recognition pipeline, the decoder reconstructs the ordered musical symbol sequence from LSTM-enhanced Transformer encoder features. It combines outputs of the previous *m* encoder layers and the *AR* component, then applies *ReLU* activation followed by a linear layer to generate predictions in Eq. (19)

$$\hat{z}_{g+1} = \text{Linear}(\text{ReLU}(Q_1, Q_2, \dots, Q_m, Q_{AR})) \quad (19)$$

Here, $\hat{z}_{g+1}$ is the predicted musical symbol at the next time step, representing the decoder output for sequence reconstruction, and Q_{AR} is the output of the autoregressive (AR) component. It represents the contribution of previously predicted musical symbols or temporal context in the sequence. The proposed music-score recognition method helps the decoder maintain rhythmic continuity, melodic progression, and temporal alignment by incorporating historical note information into the prediction of the next symbol. This ensures that the decoder generates structured, temporally consistent, and accurate musical notation, achieving the reliable piano sheet-music transcription

into a digital, editable format. Figure 3 shows the TCMN model for converting piano sheet-music images into alignment-free symbolic notation.

3.2.9 The Locust Swarm Optimization Algorithm (LSA) is used for adaptive optimization of model parameters and feature selection

The LSA optimizes critical training and preprocessing hyperparameters that directly affect music-score transcription accuracy. These include the learning rate, CNN kernel selection, Transformer embedding dimension, attention head scaling, symbol-segmentation thresholds, and CTC decoding weights. Optimizing these interdependent parameters jointly improves convergence stability and reduces sequence-level transcription errors.

The population of locust agents represents candidate solutions in the multi-dimensional optimization domain. Let y_j^l denote the current solution vector of the j^{th} locust in generation l . Each candidate solution corresponds to a potential combination of image-processing features or neural-network hyperparameters. y_j^{l+1} is the updated position of the j^{th} agent at iteration $l + 1$ is expressed in Eq. (20)

$$y_j^{l+1} = y_j^l + \Delta y_j \quad (20)$$

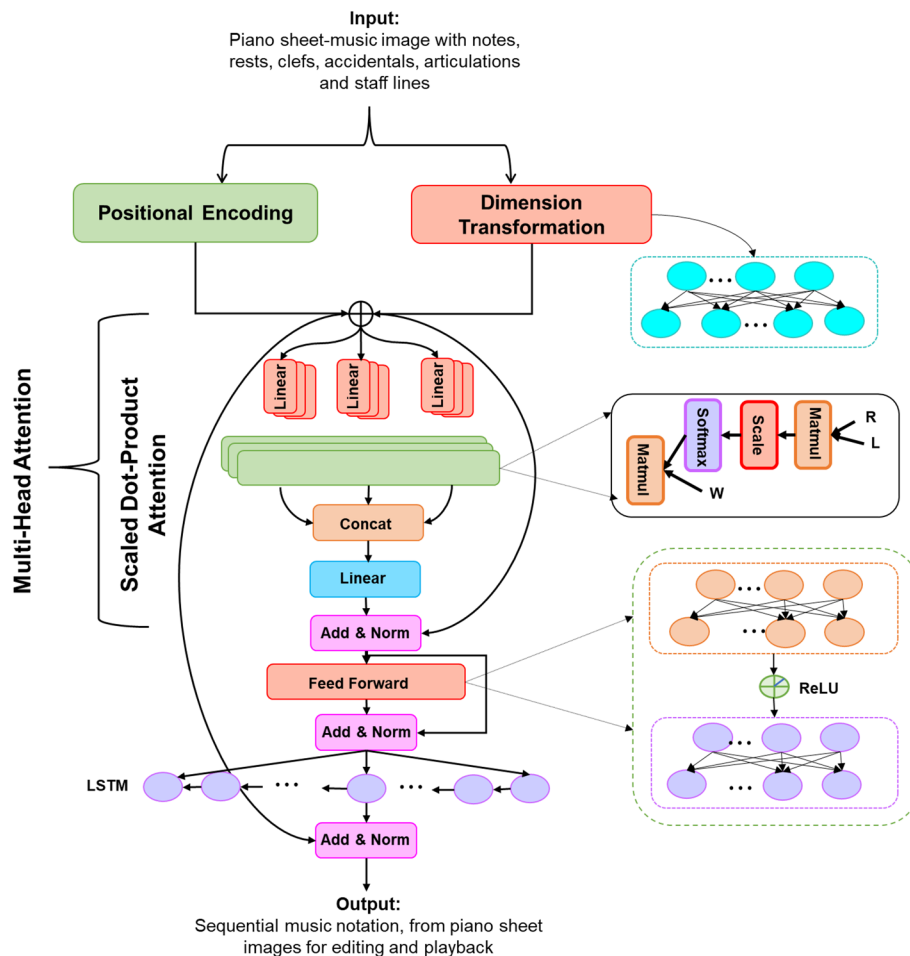


Fig. 3 Transformer-based convo memory network architecture

Where, Δy_j representing the movement step of candidate j influenced by swarm interaction, y denotes the input feature vector derived from musical symbols, and y_j is the feature component corresponding to the j^{th} musical element. The attraction–repulsion mechanism governing swarm topology is modelled through Eq. (21)

$$t(s) = Kar.e^{\frac{s}{lim}} - e^{-s} \quad (21)$$

Where, $t(s)$ denotes the interaction function influencing movement, s is the Euclidean distance between two agents, Kar is the attraction/repulsion coefficient, and e^{lim} represents the boundary constraints applied during position update, e^{-s} is the exponential component shaping repulsion-attraction dynamics. The force exerted by the k^{th} on the j^{th} locust is expressed in Eq. (22)

$$t_{jk} = t(s_{jk})c_{jk} \quad (22)$$

Where, $s_{jk} = |y_k - y_j|$ and $c_{jk} = (y_k - y_j)/s_{jk}$ is the normalized directional vector, t_{jk} represents the similarity score between symbol descriptors, s_{jk} denotes the relative spatial relation between two notational elements, c_{jk} is the distance between the j^{th} and k^{th} symbol vectors, y_k denotes the parameter vector of the k^{th} candidate solution influencing y_j , and Indices j, k represent symbol or candidate-solution identifier. The cumulative swarm influence acting on the j -th locust in Eq. (23)

$$T_j = \sum_{k=1, k \neq j}^M t_{jk} \quad (23)$$

Where T_j is the updated state of agent j after interaction, M is the number of candidate solutions, T is the global search process during optimization, and yielding the positional change (Δy_j) in Eq. (24)

$$\Delta y_j = T_j \quad (24)$$

Each candidate solution is evaluated using a food-index fitness function, where higher values correspond to superior recognition performance. The multi-dimensional search space is defined as Eq. (25)

$$T = \{m^l \in S^o | low_e \leq m^l \leq upp_e\} \quad (25)$$

Here, m^l is the step length factor at iteration l (maximum number of iterations allowed), S^o denotes an o -dimensional continuous search space used for parameter tuning, low_e and upp_e specifying lower and upper bounds of each optimization dimension. LSA is applied during the training phase as an outer optimization loop. Each locust agent encodes a candidate parameter configuration, and the TCMN model is trained for a fixed number of epochs using that configuration. Model accuracy and F1-score are used as fitness objectives, after which the best configuration is selected for final training. LSA does not affect inference.

Many traditional optimization strategies assume a smooth objective function and struggle with the highly non-convex, mixed discrete–continuous parameter space of deep music-recognition models. In contrast, LSA effectively balances exploration and exploitation, enabling efficient global search and robustness against local optima caused by noisy gradients and symbol-density variations.

3.2.10 Clarification of LSA implementation and convergence

The LSA optimizes the TCMN model by minimizing a weighted fitness function combining transcription errors shown in Eq. (26):

$$F = \alpha \cdot CER + \beta \cdot SER + \gamma \cdot LER, \quad \alpha + \beta + \gamma = 1 \quad (26)$$

The objective is based on a weighted sum of Character Error Rate (CER), Symbol Error Rate (SER), and Line Error Rate (LER). The weighting coefficients strike a balance between the accuracy of the symbols and the order in the process of optimisation. Reduction of this goal guides the LSA to parameter sets that maximise the transcription reliability. The condition $\alpha + \beta + \gamma = 1$ ensures that the three error terms (CER, SER, and LER) are properly normalized. It prevents any single metric from dominating the optimization process.

The attraction-repulsion coefficient is dynamically planned and uses greater repulsion at the beginning of the iterations to stimulate exploration and a gradual increase of attraction towards the end to stimulate convergence. Convergence is achieved by limited parameter changes and fitness-based swarm contractions, and empirical monotonic improvement has been reached as the iterations increase, which demonstrates a stable and trusted optimisation procedure.

When compared with other traditional optimization methods, LSA consistently yields better optimization behavior and improved transcription performance. The LSA leads to more effective parameter tuning and reduced recognition errors, highlighting its advantage over conventional optimization approaches.

3.2.11 Connectionist temporal classification (CTC) for alignment-free music-score transcription

In the piano sheet-music recognition framework, the Transformer-based decoder generates a sequential digital output from the LSTM-enhanced encoded features. To ensure alignment-free transcription of variable-length visual sequences into symbolic notation, the system incorporates CTC. To predict the most likely label sequence $\hat{t}$ for each input image slice y derived from the piano score is represented in Eq. (27).

$$\hat{b} = \underset{t \in \Sigma_b}{\operatorname{argmax}} Q(t|y) \quad (27)$$

Here, $\hat{b}$ is the predicted alignment sequence produced by the CTC decoder, arg is the argument of the maximization function, $\max_{t \in \Sigma_b}$ is the maximization over all valid musical symbol sequences, t is the target musical symbol sequences, Σ_b denotes the set of all possible musical notation symbols, and $Q(t|y)$ denotes the probability of symbol sequences t given input y . CTC approximates this probability by summing over all possible label alignments $B_{t,y}$ across U frames in Eq. (28)

$$Q(t|y) = \sum_{b \in B_{t,y}} \prod_{u=1}^U q_u(b_u|y) \quad (28)$$

Here, U denotes the total number of output time frames, u is the current time frame index, $\prod$ is the product operator, $B_{t,y}$ represents all valid alignment paths for sequence t and input y , q_u is the model's probability distribution at time u , and b_u is the symbol predicted at time u . A blank character ϵ is added to handle repeated or missing

symbols, defining the extended vocabulary Σ' . The final predicted musical sequence $\hat{t}$ is obtained by collapsing repeats and removing blanks, and is defined as Eqs. (29–30)

$$\hat{t} = A(\hat{b}) \quad (29)$$

$$\hat{b} = \underset{b \in \Sigma'}{\operatorname{argmax}} \prod_{u=1}^U Q(b_u | y). \hat{t} = A(\hat{b}) \quad (30)$$

Where, A is the CTC collapse function that removes repeats and blanks, CTC ensures that the system aligns variable-length visual inputs with symbol outputs, enabling accurate, sequential, and structured digital transcription of piano sheet music.

The proposed TCMN-LSA combines several stages in order to improve the accuracy, robustness, and efficiency of automatic piano sheet-music recognition. The CNN module identifies spatial patterns on the basis of multi-kernel filters to identify the geometry of the staff, note-head shapes, and symbol strokes. Transformer encoder captures long-range symbol dependencies with positional encoding, multi-head attention, residual connections, as well as feed-forward layers. The LSTM maintains the continuity in sequence through forget, input, and output gates. Transformer decoder executes autoregressive reconstruction of symbols, and LSA executes the task of optimizing feature weights, attention dimensions, and segmentation thresholds. Lastly, CTC offers a non-alignment transcription, which produces high-quality, structured, and accurate digital music notation to facilitate precise transcription, editing, and archiving, as well as facilitating high-level music analysis and digital preservation. The formulated TCMN-LSA model is described in Algorithm 1.

3.2.12 CTC decoding strategy and temporal slicing policy

The proposed TCMN-LSA framework uses greedy CTC decoding, where the most probable symbol is selected at each time step, followed by collapsing repeated labels and removing blank tokens to produce the final transcription. Beam search decoding and external language models are not employed to avoid additional computational overhead and ensure stable, real-time inference. For temporal processing, piano sheet images are divided using a sliding-window slicing policy with fixed-width vertical slices, partial horizontal overlap, and left-to-right alignment to preserve symbol continuity and accurate sequential ordering.

Algorithm 1 TCMN-LSA model.

Input: Sheet music images I
Output: Predicted musical symbol sequence S

Step 1: Initialize TCMN Architecture
Initialize CNN layers for spatial feature extraction
Initialize the Transformer encoder and decoder
Initialize LSTM for temporal sequence modeling
Initialize CTC decoder for alignment-free transcription

Step 2: Initialize Locust Swarm Algorithm (LSA)
Initialize the population of locust agents
Initialize attraction and repulsion coefficients
Initialize fitness function based on recognition accuracy and CER
Set the maximum number of iterations

Step 3: Training and Optimization Process
For each training epoch, do
 Extract spatial features using CNN
 Encode contextual dependencies using Transformer
 Model temporal dependencies using LSTM
 Decode symbol sequence using Transformer decoder + CTC
 If LSA optimization is enabled, then
 Evaluate the fitness of each locust agent
 Update agent positions using the attraction–repulsion mechanism
 Select the best parameter configuration
 Update TCMN parameters using LSA
 Else
 Update TCMN parameters using standard backpropagation
 End If
End For

Step 4: Inference
Input test image into the trained TCMN-LSA model
Generate alignment-free musical symbol sequence S using CTC
Return S

4 Performance evaluation

A benchmarked music-score recognition dataset (GrandStaff and Camera-GrandStaff) containing staff-line structures, noteheads, rests, clefs, accidentals, beams, and camera-induced distortions is used to evaluate the proposed TCMN-LSA model. The data include key parameters such as symbol density, staff-spacing variation, lighting noise, and resolution differences. Both datasets are split into 70% training, 15% validation, and 15% testing. The model is implemented in Python using TensorFlow and PyTorch, and its performance is assessed using accuracy, precision, recall, F1-score, CER, SER, LER, and CER_{bug}, covering symbol-wise and sequence-level transcription quality. Table 2 shows the hyperparameters of the proposed TCMN-LSA music-score recognition model.

4.1 Computational cost and resource analysis

Computational Cost and Resource Analysis evaluates the hardware and time requirements of a model during training and inference. It includes metrics such as model size, training time, inference latency, and memory usage. This analysis assists in assessing the efficiency, scalability, and practical deployability of the proposed approach.

The computational efficiency of the proposed TCMN-LSA model, with about 50 million parameters, is shown in Table 3. The total time to train takes 30 epochs with an average duration of about ~2 h on a Tesla V100 GPU and uses about 16GB of GPU memory

Table 2 Hyperparameters of modules in the proposed music-score recognition model

Module	Hyperparameters
CNN	Conv layers 3–5, Kernel sizes $3 \times 3, 5 \times 5, 7 \times 7$, Filters 64–256, Activation ReLU, Pooling Max 2×2 , Stride 1–2
Transformer Encoder	Layers 4–6, Heads 8, Embedding dim 256–512, Feed-forward 1024, Dropout 0.1–0.3
LSTM	Hidden units 256–512, Layers 1–2, Dropout 0.2, Gate activation Sigmoid/Tanh
Transformer Decoder	Layers 2–4, Heads 8, Output dim 256
AR Component	Context window previous m symbols
CTC	Blank token Yes, Loss CTC Loss
LSA	Population 20–50, Step size adaptive, Iterations 50–100, Attraction/repulsion coefficient adaptive, Parameter bounds lower/upper, Fitness Recognition accuracy/F1
Training Reproducibility	Random seed for training: 42 ; Random seed for data shuffling: 24 ; Random seed for weight initialization: 15
Conv layers	5
Filters	128
Batch size	32
Optimizer	Adam, with a learning rate of 0.001, beta1 = 0.9, beta2 = 0.999
Learning rate	0.001
Epochs	30
Early stopping:	Applied with a patience of 5 epochs based on validation loss
Weight initialization	Xavier initialization for CNN layers, and normal initialization for Transformer and LSTM layers

Table 3 Computational cost analysis of the proposed TCMN-LSA model

Metric	Value
Model size	~ 50 million parameters
Training epochs	30
Training time	~ 2 h per epoch (Tesla V100 GPU)
Inference latency	~ 0.2 s per image (batch size = 1)
GPU memory usage	~ 16 GB (batch size = 32)

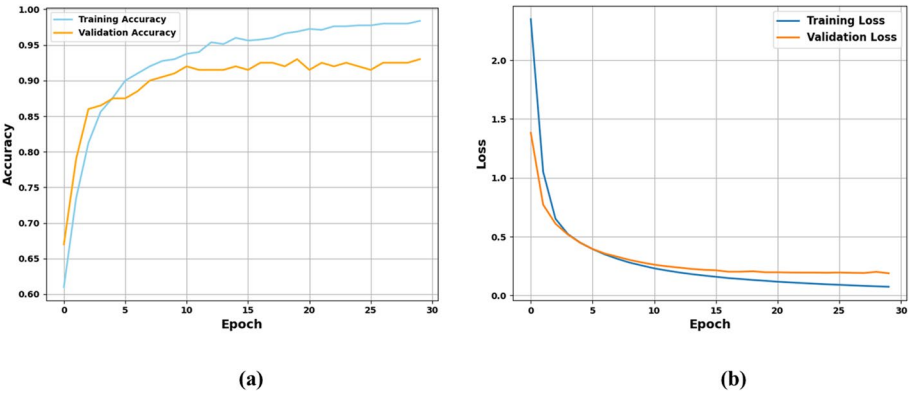


Fig. 4 Training vs. validation performance of (a) accuracy and (b) loss of the proposed TCMN-LAS model

with a batch size of 32. The model is able to attain a low latency of about 0.2 s per image; thus, it can be easily deployed in practice.

4.2 Performance analysis

As the experiment shows, the proposed TCMN-LSA model effectively recognizes piano sheet music. Figure 4(a) shows training vs. validation accuracy and loss over 30 epochs. Training accuracy steadily rises from ~0.61 to ~0.99, validation accuracy from ~0.67 to

~0.93, while Fig. 4(b) training loss drops from ~2.3 to ~0.08 and validation loss from ~1.38 to ~0.19, indicating effective learning, minimal over-fitting, and robust performance of the TCMN-LSA model.

4.3 Evaluation metrics

The principal measurement indicators of the effectiveness of the TCMN-LSA model in music-score recognition were precision, recall, accuracy, CER, SER, LER, and CER_{bug} . These indicators were used to measure the symbol recognition, sequence accuracy, and alignment.

Precision: The score in which the musical symbols are accurately predicted in comparison to all symbols predicted, which shows that the model can avoid false positives in the transcription of piano sheet music. *Recall*: Represents the ability to identify predicted symbols correctly among total actual symbols, an evaluation of the ability of the system to represent all notes, rests, and articulations in the score. *Accuracy*: It gives the ratio of correctly recognized musical symbols to total symbols, the measure of overall effectiveness of the TCMN-LSA model in structured digital music transcription. *Character Error Rate (CER)*: Percentage of incorrect predictions of individual musical symbols (notes, rests, accidentals) against all and only musical symbols, and metric of fine-grained error in transcription of piano scores. *Symbol Error Rate (SER)*: Fraction of symbols of music misrecognized in sequence, and measures the accuracy of the sequence, and the contextual performance of transcription of the score. *Line Error Rate (LER)*: Measures error in predicted labels on the notes relative to ground truth, which includes insertions, deletions, and substitutions, emphasizing misalignment or misclassification. CER_{bug} : A specialized version of CER that identifies repeated or missed symbols due to alignment error or decoding error, which evaluates the strength of the CTC-based alignment in music-score recognition.

Onset F1: Onset F1 evaluates how precisely and completely the model detects note start times. It is the harmonic mean of onset precision and recall. This metric reflects temporal accuracy in note activation. *Note F1*: Corresponds to the accuracy of predicted notes, which refers to the accuracy of the notes by their pitch and duration. This measure is a trade-off between accuracy and recall at the note level; the higher the value, the more accurate note identification and duration modelling.

AUC (Area Under the Curve) Denotes the capacity of the model to differentiate between bad and good predictions at different thresholds. It provides a performance summary across all boundaries of decisions; the more the AUC, the greater the overall discriminative power.

4.4 Analysis of the proposed TCMN-LSA model's performance

The confusion matrix shows the performance of the classification system in terms of the comparison of the predicted music symbols and ground-truth symbols. The expected classes are displayed on the horizontal axis, whereas the actual classes are displayed on the vertical axis. The number of samples that are assigned to a particular predicted-true label combination is noted in each entry of the matrix, whose counts vary between 7 and 70, with a total of 150 assessed instances. This representation (Fig. 5) explains correctly and incorrectly categorized symbols, explains call-made distributions, and shows

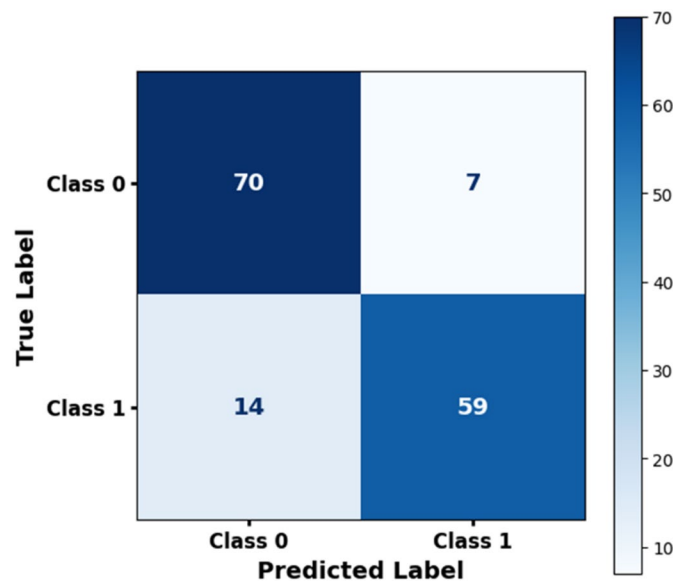


Fig. 5 Confusion matrix for TCMN-LSA model



Fig. 6 Z-score normalization of the TCMN-LSA model

the whole labeling consistency achieved by the TCMN-LSA model. It thus provides an accurate description of symbol-level recognition processes in the tested piano sheet-music data and enables a detailed evaluation of high-accuracy areas and consistency of misclassification. The visualization of the notes over time and pitch is represented by the piano-roll-style view. Time progression is represented in the horizontal axis, and pitch over the various octaves is depicted in the vertical axis; every point on the axis represents a musical note.

To get an additional insight into the recognition behaviour of the proposed model, an error analysis was performed based on the confusion matrix presented in Fig. 5. Testing was conducted on the samples of the GrandStaff dataset that contained various musical symbols, including noteheads, rests, clefs, accidentals, beams, and bar lines in polyphonic piano scores. The majority of the predictions were lying on the diagonal, which means that most symbols were classified correctly. But noteheads, rests, accidentals, beams, and clefs showed more errors. Intersections of staff lines and low contrast obscure boundaries were sometimes confused with rests. Similar strokes sometimes led to the accidentals being mistaken for beam fragments or thin staff artefacts. There were small misclassifications due to clefs since the shape was distorted by noise or image distortion. Mistakes were primarily found in high-density polyphonic texts or marginally damaged images, wherein the symbols overlap, and the similarity in the picture decreases the separability. The matrix validated high consistency in recognition with residual errors based on structural similarity and not on model constraints. As a result, Fig. 6 emphasizes the pattern of piano-score, distribution of symbols, and temporal-spatial dependence of the TCMN-LSA processing.

The CNN feature extraction module produces feature maps, which represent edges, nodal formations, patterns of staff and lines, among other visualization indicators of sheet-music inputs. Spatial dimensions are considered to be axes, and the strength of the activation is considered to be intensity. In this regard, Fig. 7 illustrates hierarchical patterns of space that are critical in order to have proper musical-symbol recognition.

The Receiver Operating Characteristics (ROC) curve illustrates the ability of the music-recognition model to correctly recognize musical symbols across varying decision thresholds. Axes: X-False Positive Rate (FPR), Y-True Positive Rate (TPR), Ranges: 0.0–1.0, increments of 0.2. Therefore, Fig. 8 shows the ROC curve, illustrating the TCMN-LSA diagnostic ability, the trade-off between true positive and false positive rates, and overall classification performance across different thresholds.

The Precision-Recall curve plots Precision versus Recall within a 0–1 range, with an AP of 0.98 indicating highly effective music-symbol classification. Thus, Fig. 9 demonstrates how well the TCMN-LSA model balances symbol-level precision and recall, reflecting strong overall transcription capability across varying thresholds.

The Sensitivity-Specificity shows a binary classifier's performance across a threshold, reflecting the trade-off between sensitivity and specificity. Axes & range: Y-axis: sensitivity (0–1), X-axis: specificity or FPR (0–1). Thus, Fig. 10 illustrates how effectively the TCMN-LSA model distinguishes correct and incorrect piano-music symbol predictions, demonstrating its reliability in sheet music recognition.

The Multiple performance metrics on axes radiating from a central point. It displays NEA, VAA, TSRA, RRA, and PRA values ranging from 80 to 98%, representing the model's capability across diverse recognition tasks. Hence, Fig. 11 presents the radar chart illustrating the TCMN-LSA model's overall effectiveness in achieving accurate and reliable piano sheet music recognition.

4.5 Comparative analysis

The existing research on music transcription has demonstrated several approaches that have been applied to enhance symbol recognition, sequential modelling, and polyphonic score transcription, such as Non-negative Matrix Factorization (NMF) [28], which is applied to separate sources in polyphonic music, allowing the recognition of individual voices, CNN with a Transformer Encoder (CNNT) [29] applied Bekern encoding to

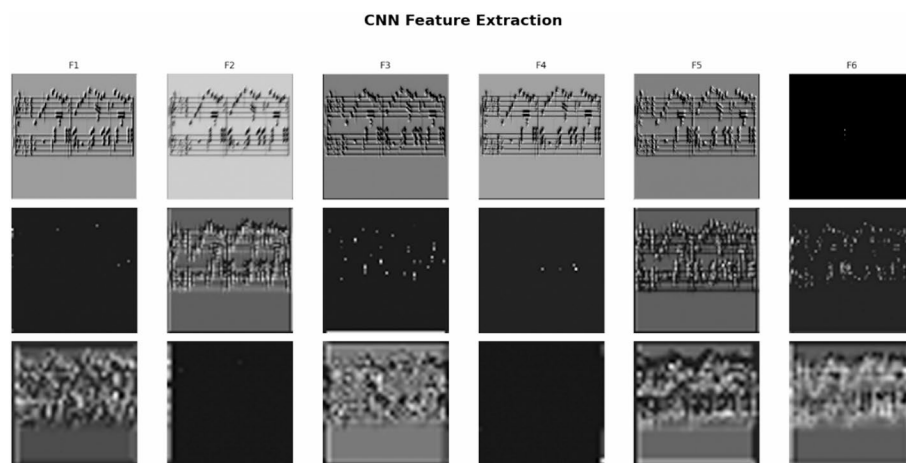


Fig. 7 CNN feature extraction of TCMN-LSA model

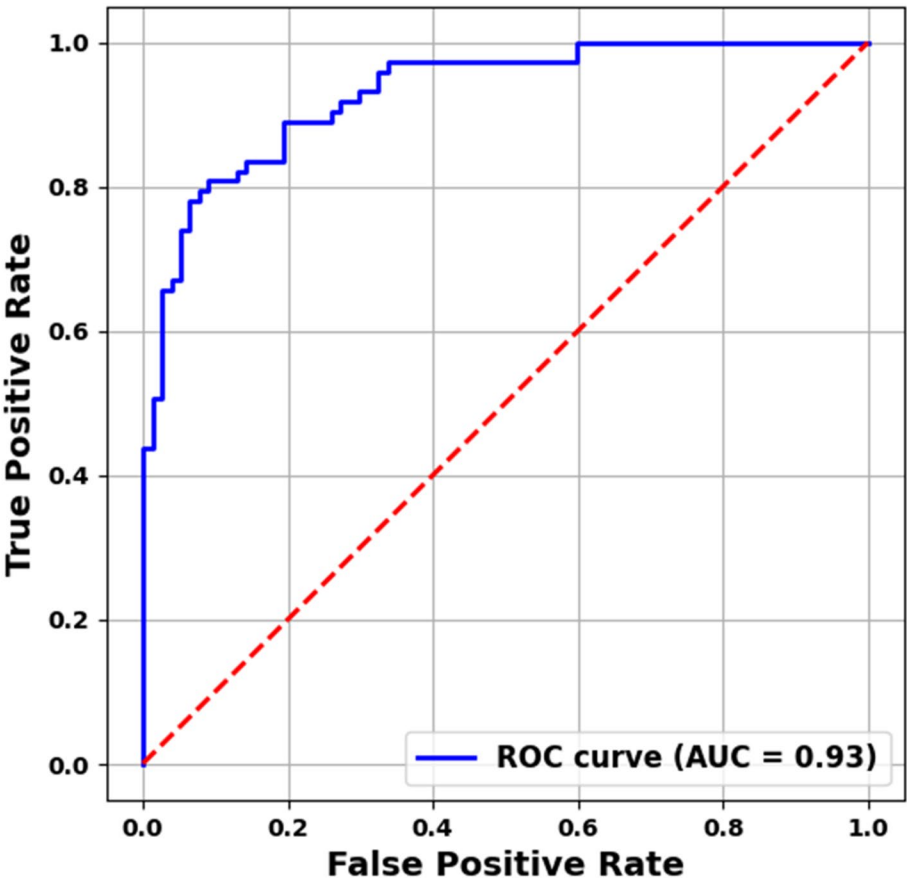


Fig. 8 ROC curve of the TCMN-LSA model

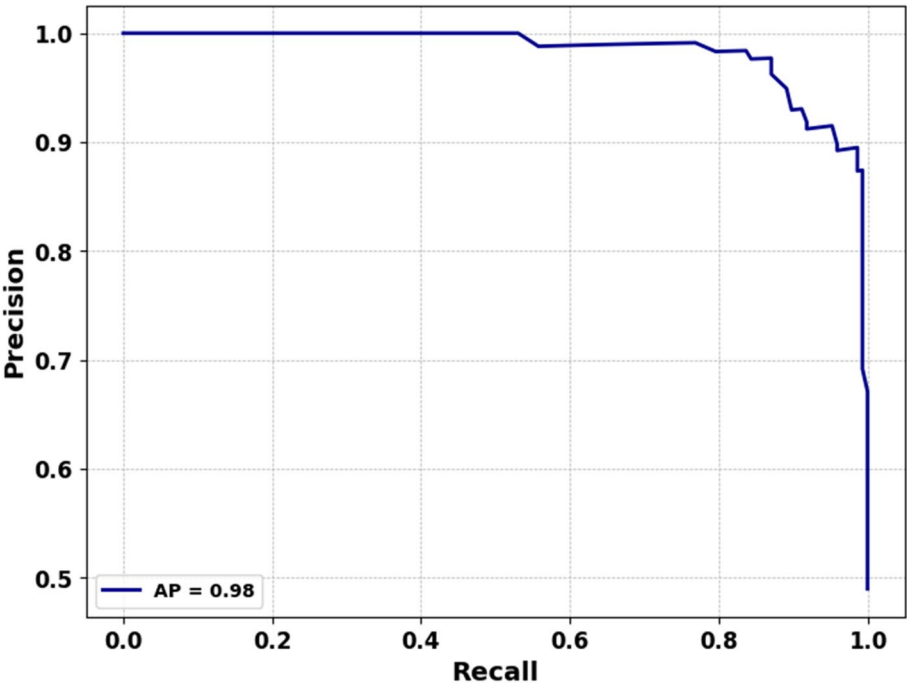


Fig. 9 Precision-Recall curve of the TCMN-LSA model

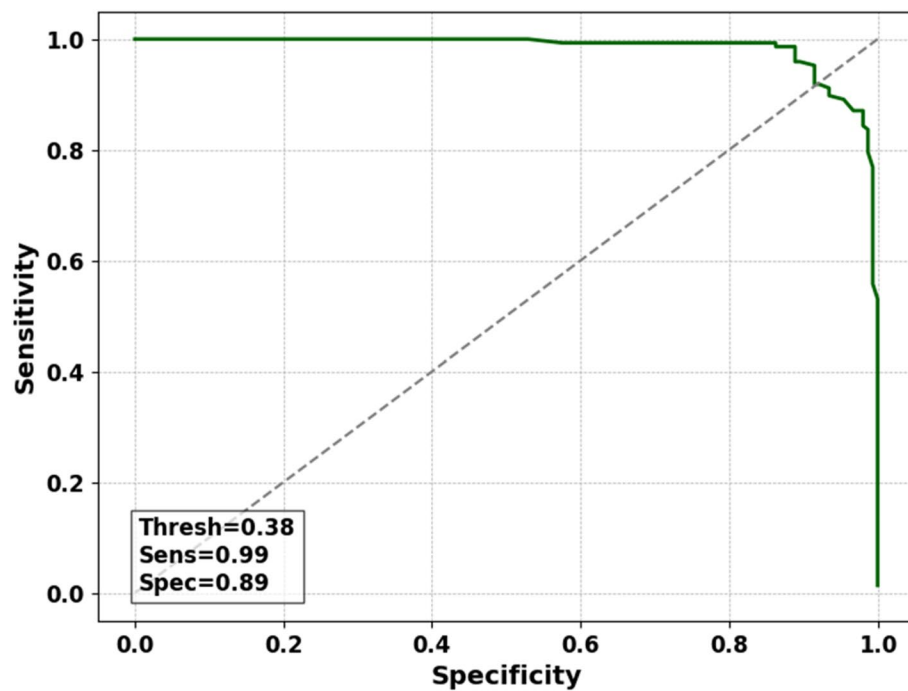


Fig. 10 ROC curve for TCMN-LSA model

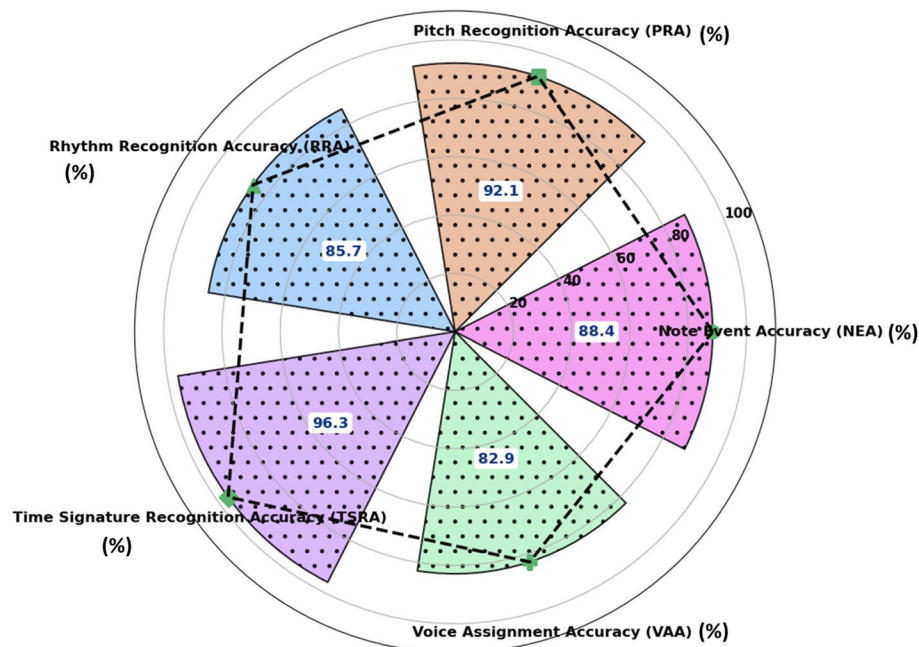
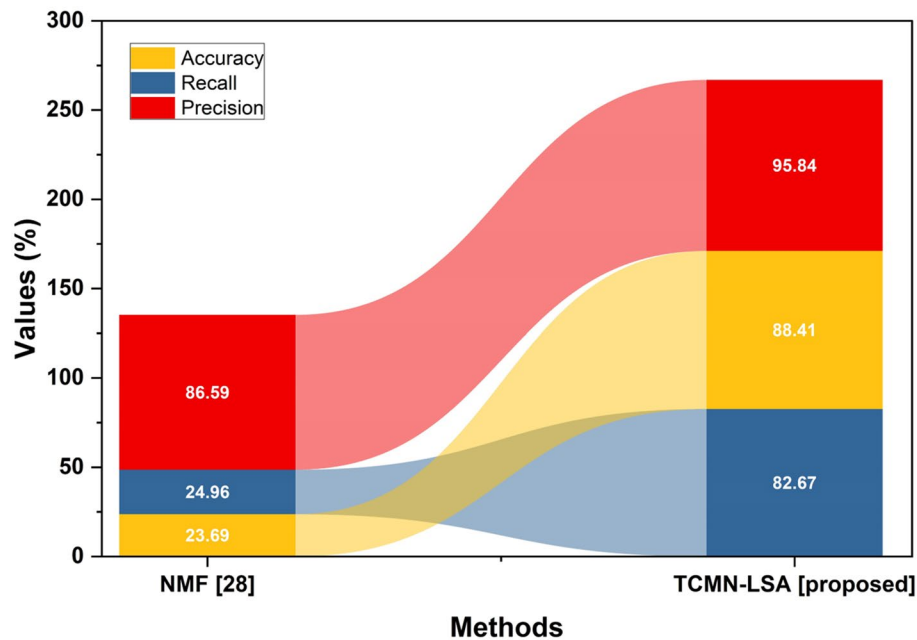


Fig. 11 Multimetric performance of the TCMN-LSA model

enhance symbol presentation in an optical music recognition setup, and Zeus [30], a Convolutional Recurrent Neural Network (CRNN)-based model, adding convolutional layers, bidirectional LSTM, attention, and CTC layer to improve polyphonic transcription. To ensure methodological fairness, state-of-the-art baseline models, including CNNT [29] and Zeus [30], were reimplemented and evaluated on the same datasets under identical experimental conditions. This enables direct and unbiased comparison

Table 4 Performance comparison of NMF and the proposed TCMN-LSA model in Music score Recognition

Methods	Metrics (%)		
	Precision	Recall	Accuracy
NMF [28]	86.59	24.96	23.69
TCMN-LSA [proposed]	95.84	82.67	88.41

**Fig. 12** Evaluation metrics compared with NMF and TCMN-LSA

between the proposed TCMN-LSA model and existing approaches for automatic music score recognition.

Performance comparison was conducted using the MAESTRO Dataset and the MAPS Dataset, both of which are publicly available benchmark datasets widely used in music transcription research. Using open datasets will guarantee equal comparison, reproducibility, and open evaluation of the proposed method of TCMN-LSA (GrandStaff Dataset and Camera GrandStaff Dataset) in relation to DPPM [31].

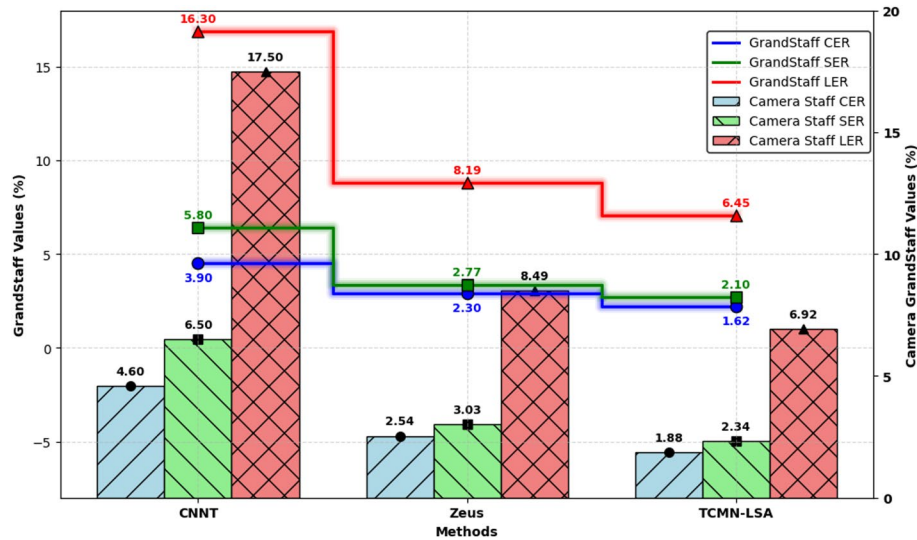
To make a reasonable and repeatable comparison, the baseline models CNNT [29] and Zeus [30] were re-implemented according to their original architecture and training setups. The proposed TCMN-LSA model was trained and tested on the same datasets of GrandStaff and Camera-GrandStaff with the same preprocessing processes and evaluation measurements as the two baselines. This experimentally controlled environment ensures the valid and significant performance comparison of all the models.

These approaches have weaknesses and limitations such as limited polyphonic performance, the need to operate on aligned sequences, and reduced performance on distorted or handwritten scores.

Automatic music score recognition transforms printed or handwritten sheets into editable digital notation. According to Table 4; Fig. 12, the proposed TCMN-LSA model outperforms the existing NMF [28] method in all metrics, achieving 95.84% of precision,

Table 5 Comparative evaluation of existing and proposed models on GrandStaff and Camera-GrandStaff datasets

Methods	GrandStaff			Camera GrandStaff		
	CER	SER	LER	CER	SER	LER
CNNT [29]	3.9	5.8	16.3	4.6	6.5	17.5
Zeus [30]	2.30	2.77	8.19	2.54	3.03	8.49
TCMN-LSA [proposed]	1.62	2.10	6.45	1.88	2.34	6.92

**Fig. 13** Comparison of CER, SER, and LER values of existing and proposed models**Table 6** CER_{bug} comparison between the existing and proposed on the GrandStaff and Camera-GrandStaff dataset

Methods	GrandStaff	Camera-GrandStaff
	CER_{bug}	CER_{bug}
Zeus [30]	1.68	1.91
TCMN-LSA [proposed]	1.21	1.38

82.67% of recall, and 88.41% of accuracy, demonstrating robust, alignment-free transcription and significant improvements in symbol recognition and reliability.

According to Table 5; Fig. 13, the proposed TCMN-LSA model demonstrates superior performance compared with existing methods, CNNT [29] and Zeus [30] across all evaluation metrics for both Grandstaff and Camera-Grandstaff datasets. It achieved lower CER, SER, and LER, with values of 1.62%, 2.10%, and 6.45% for Grandstaff, and 1.88%, 2.34%, and 6.92% for Camera Grandstaff. These results indicate enhanced accuracy, robust symbol recognition, and improved transcription reliability, directly supporting and achieving more precise and consistent piano music score recognition.

Table 6 mentions both dataset values. The proposed TCMN-LSA model outperforms Zeus [30] in CER_{bug} for both datasets. It achieves lower CER_{bug} values of 1.21% and 1.38%. Compared with 1.68% and 1.91% for Zeus, demonstrating more accurate and reliable music-score transcription and improved handling of symbol alignment and transcription errors.

Table 7 shows that TCMN-LSA consistently outperforms DPPM across datasets. On GrandStaff, it achieves 95.12% accuracy, 93.86% onset F1, 94.21% note F1, and 96.48%

Table 7 Performance comparison of DPPM and the proposed TCMN-LSA model on the MAESTRO and MAPS datasets

Metrics	DPPM [31]		TCMN-LSA [proposed]	
	MAESTRO Dataset	MAPS Dataset	GrandStaff Dataset	Camera GrandStaff Dataset
Accuracy	92.35 ± 0.03	89.65 ± 0.02	95.12 ± 0.02	92.34 ± 0.02
Onset F1	90.50 ± 0.02	87.70 ± 0.02	93.86 ± 0.02	90.92 ± 0.02
Note F1	90.87 ± 0.03	88.55 ± 0.02	94.21 ± 0.02	91.58 ± 0.02
AUC	93.15 ± 0.02	94.45 ± 0.02	96.48 ± 0.01	97.02 ± 0.01

Table 8 Music-score recognition performance comparison using combined GrandStaff datasets

Methods	Accuracy (%)	Precision (%)	Recall (%)
NMF [28]	68.73	74.21	58.34
CNNT [29]	79.52	85.16	72.48
Zeus [30]	83.67	89.43	77.91
DPPM [31]	86.24	92.08	80.36
TCMN-LSA [proposed]	88.41	95.84	82.67

Table 9 Statistical significance analysis of model performance over three independent runs

Model	Mean accuracy (%)	SD	t-value	p-value	Significance
LSA	76.12	0.71	9.43	< 0.001	Significant
TCMN	83.47	0.62	15.82	< 0.001	Significant
TCMN-LSA (Proposed)	88.41	0.48	23.91	< 0.001	Significant

AUC. On Camera-GrandStaff, performance remains strong with 92.34% accuracy and 97.02% AUC, demonstrating robust generalization.

The comparative performance analysis of NMF [28], CNNT [29], Zeus CRNN model [30], DPPM [31], and the proposed TCMN-LSA model was conducted using a combined GrandStaff and Camera-GrandStaff dataset for automatic music-score recognition evaluation. Detailed comparative results appear in Table 8.

The proposed TCMN-LSA model records the best results of 88.41% accuracy, 95.84% precision, and 82.67% recall, indicating a higher musical-symbol recognition and piano sheet-music transcription ability than conventional models.

Paired t-test

A paired t-test is a statistical test that is applied to compare two related samples of means. It determines the significance of the differences observed. It is actively utilized when the models are tested on the same data more than once. In order to implement statistical reliability and reproducibility, the proposed TCMN-LSA model was trained and tested five times using varying random seeds and keeping all hyperparameters fixed.

Table 9 indicates that the mean accuracy of TCMN3SA (88.41%) and its variability (SD = 0.48) are the highest. The significant improvement is statistically significant with a large t-value of 23.91 and $p < 0.001$. Comparatively, TCMN is recorded to have a higher level of performance, 83.47% (SD = 0.62), and LSA is recorded at 76.12% (SD = 0.71), indicating lower performance.

Ablation study

An ablation experimentally manipulates or deletes the parts of a model. It assesses the performance effect of all the components in isolation. The performance deterioration justifies the need for such a component.

Table 10 Ablation analysis of model configurations for music score recognition performance

Configuration	Accuracy (%)	Precision (%)	Recall (%)
CNN only	72.84	81.37	64.52
CNN + Transformer	78.96	87.05	71.88
CNN + LSTM	77.41	85.62	70.11
TCMN without LSA	85.02	92.31	78.44
TCMN-LSA without CTC	84.19	91.76	77.83
TCMN-LSA (Proposed) with CTC	88.41	95.84	82.67

Table 11 5-Fold cross-validation results of the proposed TCMN-LSA model on the GrandStaff dataset

Fold	Accuracy (%)	Precision (%)	Recall (%)	CER (%)	SER (%)	LER (%)
Fold-1	88.12	95.60	82.21	1.68	2.15	6.51
Fold-2	88.47	95.91	82.54	1.61	2.09	6.44
Fold-3	88.69	96.10	83.01	1.59	2.04	6.39
Fold-4	88.28	95.72	82.36	1.64	2.12	6.48
Fold-5	88.49	95.89	82.75	1.58	2.08	6.42
Mean $\pm$ Std	88.41 $\pm$ 0.21	95.84 $\pm$ 0.18	82.67 $\pm$ 0.29	1.62 $\pm$ 0.04	2.10 $\pm$ 0.04	6.45 $\pm$ 0.05

Table 10 presents ablation results across model configurations. CNN-only achieves 72.84% accuracy, improving with CNN+Transformer (78.96%) and CNN+LSTM (77.41%). The TCMN without LSA achieves an accuracy of 85.02%, and the TCMN-LSA without CTC slightly decreases performance to 84.19%. The greater change when CTC is removed is because the CTC layer directly controls sequence alignment of the predicted symbols and ground-truth notation in transcription. Even though the effect of LSA removal is a little smaller, LSA enhances the parameter optimization and stability of convergence during training. The combination of the two components thus allows the model to attain the maximum overall recognition performance. The proposed TCMN-LSA with CTC provides the highest accuracy of 88.41%, precision 95.84%, and recall 82.67%, which validates the contribution of each module.

5-Fold cross-validation

A 5-fold cross-validation process splits the dataset into five equal parts. Each iteration has four folds used to train the model, and the rest of the fold is used to test the model. This procedure is done five times to secure a stable and non-discriminating performance appraisal.

Table 11 reports 5-fold cross-validation results of the proposed TCMN-LSA model. Consistent accuracy around 88.41% with low standard deviation (± 0.21) and minimal variation in CER ($1.62\% \pm 0.04$) confirms strong generalization, robustness, and reliability across different data partitions.

4.6 Discussion

An efficient DL-based system is introduced to accurately and alignment-free transcription of multifaceted piano sheet music on different and natural datasets. Current methods have notable limitations, even though they are effective. Non-negative matrix factorisation (NMF) is not useful with noisy and handwritten scores and cannot promote contextual modelling of complex layouts [28]. Transformers based on a convolutional neural network (CNN) perform better in representing symbols, but worse with polyphonic and misaligned sequences [29]. Zeus is more effective in temporal modelling,

but at a very high computational cost and reduced sensitivity to distorted inputs [30]. De-convolutional probabilistic models (DPPM) are also good at controlled datasets, but with low generalisation to different recordings of real-life pianos [31].

The identification of music scores, which is founded on a deep-learning model, was implemented using images of full scores, which generated a direct result to predict the onset times and pitches of notes [32]. The accuracy of the experiment had a score of 0.95 for timing and 0.97 for pitch. However, traditionally adopted approaches of SVM were less effective, and recognition was sensitive to complex structures of scores and data heterogeneity. Recognition of music using a CRNN architecture with the addition of the residual depth-wise separable convolution, squeeze excitation attention modules, which resulted in lower note-recognition and sequence-error rates and reduced recognition time [33]. Their drawback was dependence on a specific architectural construction and sensitivity to manipulation of model parameters, which affected the generalisation of a very large range of music scores.

The suggested TCMN -LSA model addresses these drawbacks by combining CNN -based spatial feature extraction, Transformer -LSTM sequential modelling, and adaptive Locust Swarm Optimisation to ensure consistent convergence. CTC decoder makes alignment-free transcription feasible, which enhances the ability to accommodate polyphonic, distorted, and real-world scores. Comparison in GrandStaff, Camera-GrandStaff, MAESTRO, and MAPS datasets proves better generalisation, accuracy, and transcription reliability under different musical conditions.

LSA-optimised training is better in transcription performance compared to the ordinary optimisation strategies. The TCMN basic model has an accuracy of 83.47% and a CER of 2.58%, and the LSA-optimised TCMN-LSA model has an accuracy of 88.41% and a CER of 1.62%. This shows that LSA offers a more stable convergence with better sequence-level modelling compared to standard optimisation methods.

4.7 Impact of LSA

The LSA is also significant in the improvement of recognition accuracy because it allows population-based optimisation of adaptive approaches in contrast to traditional gradient-based approaches. In comparison to other optimisers who can find sub-optimal minima in complicated loss landscapes, LSA has a balanced exploration-exploitation pattern, causing more stable parameter convergence. As seen by the results of ablation, the combination of LSA and TCMN achieves better accuracy from 83.47% to 88.41% and reduces CER from 2.58% to 1.62%, with similar or better decreases in LER and SER. These advances indicate the capability of LSA to make better feature representations and time-based dependencies, which translate to higher stability and resilience in the transcription of complex and polyphonic scores of music.

5 Conclusion

To concentrate on the improvement of music-score recognition with the help of the deep-learning-based method, the images of piano sheet music of the GrandStaff and Camera-GrandStaff datasets containing printed and camera-captured scores are used. Images undergo pre-processing in terms of Z-score normalisation and Wiener filter to normalise pixel intensity and eliminate noise. The proposed TCMN-LSA method combines a multi-kernel CNN for feature extraction, a Transformer-LSTM framework for

capturing temporal and contextual dependencies, and LSA for hyper-parameter optimization. Sequential decoding is performed using a Transformer decoder with CTC alignment. Experimental results show high performance, achieving 95.84% of precision, 82.67% of recall, 88.41% of accuracy, 1.62% of CER, 2.10% of SER, 6.45% of LER, and 1.21% of CER_{bug} on GrandStaff, and 1.88% of CER, 2.34% of SER, 6.92% of LER, and 1.38% of CER_{bug} on Camera-GrandStaff. This demonstrates the model's robustness and effectiveness in accurate, structured digital music-score transcription.

5.1 Future scopes

The proposed approach is applied in the future to real-time digital music-score transcription, improved LSA-based hyperparameter optimization, to work with complex handwritten or damaged scores, audio score vocalization, to come up with lightweight Transformer variants to be used more broadly in digital music applications, and comparison with more advanced music-score recognition models.

Author contributions

Li. Conceptualization&SoftwareSong. Investigation&Writing—original draft preparation.

Funding

No funding project.

Data availability

All data generated or analysed during this study are included in this article.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Received: 16 December 2025 / Accepted: 10 April 2026

Published online: 29 April 2026

References

1. Henkel F, Widmer G. Real-time music following in score sheet images via multi-resolution prediction. *Front Comput Sci*. 2021;3:718340. <https://doi.org/10.3389/fcomp.2021.718340>.
2. Schneider D, Korfhage N, Mühling M, Lüttig P, Freisleben B. Automatic transcription of organ tablature music notation with deep neural networks. *Trans Int Soc Music Inform Retr*. 2021;4(1):14–28. <https://doi.org/10.5334/tismir.77>.
3. Ríos-Vila A, Esplà-Gomis M, Rizo D, Ponce de Leon PJ, Iñesta JM. Applying automatic translation for the optical music recognition's encoding step. *Appl Sci*. 2021;11(9):3890. <https://doi.org/10.3390/app11093890>.
4. Zalkow F, Mueller M. CTC-based learning of chroma features for score–audio music retrieval. *IEEE/ACM Trans Audio Speech Lang Process*. 2021;29:2957–71. <https://doi.org/10.1109/TASLP.2021.3110137>.
5. de la Fuente C, Valero-Mas JJ, Castellanos FJ, Calvo-Zaragoza J. Multimodal image and audio music transcription. *Int J Multimedia Inform Retr*. 2022;11(1):77–84. <https://doi.org/10.1007/s13735-021-00221-6>.
6. Lerch A, Knees P. Machine learning applied to music/audio signal processing. *Electronics*. 2021;10(24):3077. <https://doi.org/10.3390/electronics10243077>.
7. Borodovskaya L, Yavgildina Z, Dyganova E, Maykovskaya L, Medvedeva I. The possibilities of artificial intelligence in automatic musical transcription of the Tatar folk song. *Rast Müzikoloji Dergisi*. 2022;10(1):147–61. <https://doi.org/10.12975/rastmd.20221018>.
8. Yang D, Goutam A, Ji K, Tsai TJ. Large-scale multimodal piano music identification using marketplace fingerprinting. *Algorithms*. 2022;15(5):146. <https://doi.org/10.3390/a15050146>.
9. Nowitzki O, Engelhardt-Nowitzki C, Fiala ML, Wöber W. Optical music recognition of printed white mensural notation: Conversion to modern notation using object detection mechanisms. *Int J Humanit Comput*. 2022;16(1):33–49. <https://doi.org/10.3366/ijhac.2022.0275>.
10. Bolya M. AI-supported Processing of Handwritten Transcriptions for Hungarian Folk Songs in a Digital Environment. *ETH-NOMUSIC*. 2022;18:65–82. <https://doi.org/10.33398/2523-4846-2022-18-1-65-82>.
11. Alfaro-Contreras M, Ríos-Vila A, Valero-Mas JJ, Iñesta JM, Calvo-Zaragoza J. Decoupling music notation to improve end-to-end Optical Music Recognition. *Pattern Recognit Lett*. 2022;158:157–63. <https://doi.org/10.1016/j.patrec.2022.04.032>.

12. Calvo-Zaragoza J, Rizo D. End-to-end neural optical music recognition of monophonic scores. *Appl Sci*. 2018;8(4):606. <https://doi.org/10.3390/app8040606>.
13. Alfaro-Contreras M, Iñesta JM, Calvo-Zaragoza J. Optical music recognition for homophonic scores with neural networks and synthetic music generation. *Int J Multimedia Inform Retr*. 2023;12(1):12. <https://doi.org/10.1007/s13735-023-00278-5>.
14. Park J, Choi K, Oh S, Kim L, Park J. Note-level singing melody transcription with transformers. *Intell Data Anal*. 2023;27(6):1853–71. <https://doi.org/10.3233/IDA-227077>.
15. Liu Y, Wu R, Wu Y, Luo L, Xu W. A Stave-Aware Optical Music Recognition on Monophonic Scores for Camera-Based Scenarios. *Appl Sci*. 2023;13(16):9360. <https://doi.org/10.3390/app13169360>.
16. Yu P, Chen H. Deep Multilevel Cascade Residual Recurrent Framework (MCRR) for Sheet Music Recognition. *IEEE Access*. 2024;12:6941–60. <https://doi.org/10.1109/ACCESS.2024.3350880>.
17. Gu X, Ou L, Zeng W, Zhang J, Wong N, Wang Y. Automatic lyric transcription and automatic music transcription from multimodal singing. *ACM Trans Multimedia Comput Commun Appl*. 2024;20(7):1–29. <https://doi.org/10.1145/3651310>.
18. Hartelt A, Eipert T, Puppe F. Optical Medieval Music Recognition—A Complete Pipeline for Historic Chants. *Appl Sci* (2023–3417). 2024;14(16):1–26. <https://doi.org/10.3390/app14167355>.
19. Torras P, Biswas S, Fornés A. A unified representation framework for the evaluation of Optical Music Recognition systems. *Int J Doc Anal Recognit (IJRAR)*. 2024;27(3):379–93. <https://doi.org/10.1007/s10032-024-00485-8>.
20. Weiß C, Müller M. From music scores to audio recordings: Deep pitch-class representations for measuring tonal structures. *ACM J Comput Cult Herit*. 2024;17(3):1–19. <https://doi.org/10.1145/3659103>.
21. Rios-Vila A, Fuentes-Martinez E, Castellanos FJ. An implicit layout-aware transformer for full-page end-to-end optical music recognition. *Int J Multimedia Inform Retr*. 2025;14(4):1–14. <https://doi.org/10.1007/s13735-025-00385-5>.
22. Li M. Design and implementation of a piano audio automatic music transcription algorithm based on a convolutional neural network. *EURASIP J Audio Speech Music Process*. 2025;2025:1:26. <https://doi.org/10.1186/s13636-025-00412-7>.
23. Oliva-Bulpitt SB, Martínez-Esteso JP, Galán-Cuenca A, Castellanos FJ, Gallego AJ. Enhancing music score analysis with Monte Carlo dropout: a probabilistic approach to staff-region detection. *Int J Doc Anal Recognit (IJRAR)*. 2025;28:441–56. <https://doi.org/10.1007/s10032-025-00541-x>.
24. Paul A, Pramanik R, Malakar S, Sarkar R. An ensemble of deep transfer learning models for handwritten music symbol recognition. *Neural Comput Appl*. 2022;34(13):10409–27. <https://doi.org/10.1007/s00521-021-06629-9>.
25. Alfaro-Contreras M, Valero-Mas JJ. Exploiting the two-dimensional nature of agnostic music notation for neural optical music recognition. *Appl Sci*. 2021;11(8):3621. <https://doi.org/10.3390/app11083621>.
26. Lee M, Kim H, Moon M, Park SM. Computer-Vision-Based Advanced Optical Music Recognition System. *J Comput Theor Nanosci*. 2021;18(5):1345–51. <https://doi.org/10.1166/jctn.2021.9626>.
27. Rajesh S, Nalini NJ. Recognition of musical instruments using deep learning techniques. *Int J Inform Retr Res (IJRR)*. 2021;11(4):41–60. <https://doi.org/10.4018/IJRR.2021100103>.
28. Velazquez Lopez O, Rodriguez O, J.L. and, Suarez Guerra S. Application of auditory filter-banks in polyphonic music transcription. *Computación y Sistemas*. 2022;26(4):1421–8. <https://doi.org/10.13053/cys-26-4-4271>.
29. Rios-Vila A, Rizo D, Iñesta JM, Calvo-Zaragoza J. End-to-end optical music recognition for piano-form sheet music. *Int J Doc Anal Recognit (IJRAR)*. 2023;26(3):347–62. <https://doi.org/10.1007/s10032-023-00432-z>.
30. Mayer J, Straka M, Hajič J, Pecina P. Practical end-to-end optical music recognition for piano-form music. In the International Conference on Document Analysis and Recognition. Cham: Springer Nature Switzerland, 2024;14809, pp. 55–73. https://doi.org/10.1007/978-3-031-70552-6_4.
31. Ru T. Deep Learning-Based Automatic Piano Note Recognition and Performance Generation System. *Int J High Speed Electron Syst*. 2025;34(1):2540814. <https://doi.org/10.1142/S0129156425408149>.
32. Liang M. Music score recognition and composition application based on deep learning. *Math Probl Eng*. 2022;2022(1):2415857. <https://doi.org/10.1155/2022/2415857>.
33. Wang J. Research on optical music recognition based on an improved CRNN network and its application in piano teaching. *Int J Comput Sci Math*. 2025;22(2):176–91. <https://doi.org/10.1504/IJCSM.2025.149900>.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.